Lecture 11 Notes

Quantitative Data Analysis

Introduction

- We have got our model into a decent state
- The original version wasn't very good so we made some adjustments
- There are a few more assumptions underlying the model which we should discuss
- These assumptions relate more to the conceptual design of our analysis

Assumptions

We will consider the following assumptions (taken from Peter Kennedy's "A guide to econometrics"):

1. The model is correctly specified
2. The residuals have the same variance and are independent of one another
3. There are no exact linear relationships between independent variables
4. The independent variables are fixed and measured without error
5. The expected value of the residual is zero

Distributional assumptions of the residuals

- We have already discussed that the residuals should:
    - be normally distributed
    - have constant variance
- A further assumption is that they should be *independent*
- That is to say, knowing something about a particular realisation of the residual should not help you predict another value
- It results when there is a structure in our data not accounted for in the model
- This process starts when designing our analysis and collecting our data

Our data

- Each observation in our data is an individual

- If we read the documentation for the Quarterly Labour Force Survey, we find that households are the sampling unit and that every member of the household is interviewed
- People living in the same household may have certain things in common which are not accounted for in our model
- This may cause the residuals of members of the same household to be correlated
- This affects our ability to conduct hypothesis testing in a similar way to when we had heteroscedasticity

Dealing with the issue

- My feeling is that this won't be a big problem
- We have a large number of households are relatively few people per household
- There are several methods we can use to address the issue
- I am going to suggest a simple approach of using *cluster robust* standard errors
- Here, we assume correlated residuals within a group (which we specify) but assume independence between these groups
- We can estimate such a model using the *lm.cluster* function from the *miceadds* package

Cluster robust standard errors

- In our dataset, the household each person belongs to is identified by the *HSERIALP* variable
- We excluded this from our dataset, so we have to add it to our call to select and reload and reprocess our data
- Our select function should now look something like:

```
lfs <- select(lfs, SEX, HOURPAY, AGE, GOR9D, EMPLEN, FLEX22W6, FTPT, HIQUL22D,
              INDE07M, JOBTYP, MPNR02, MARSTA, NSECMJ20, BUSHR, AGES, HSERIALP)
```

Estimating the model

```
lfs %>%
  mutate(log_wage = log(HOURPAY)) %>%
  lm.cluster(data = .,
             log_wage ~ SEX + AGE + HIQUL22D + FTPT + JOBTYP + MPNR02 + new_MARSTA
+ INDE07M + NSECMJ20 + GOR9D,
             cluster = "HSERIALP") %>%
  summary()
```

Robust standard errors

```
## R^2= 0.4778
##
##                                                Estimate   Std. Error
## (Intercept)                                3.086969836 0.0296716506
## SEXFemale                                 -0.072477892 0.0087970859
## AGE                                        0.005483362 0.0004040354
## HIQUL22DHigher education                  -0.130678763 0.0161689068
## HIQUL22DA level or equivalent             -0.164029502 0.0111998679
## HIQUL22DGCSE A*-C or equivalent           -0.195033719 0.0127509367
## HIQUL22DOther qualitfication              -0.193790968 0.0209412890
## HIQUL22DNo Qualification                  -0.276496850 0.0216695497
## FTPTPart-time                             -0.030869597 0.0111997894
## JOBTYPNot permanent in some way           -0.028515091 0.0237512786
## MPNRO211-19                                0.011543459 0.0188647187
## MPNRO220-24                                0.014039373 0.0233277255
## MPNRO2Don't know but under 25             -0.030033657 0.0262168296
## MPNRO225-49                                0.013993771 0.0159845052
## MPNRO250-249                               0.047566994 0.0143197227
## MPNRO2250-499                              0.093715926 0.0194921359
## MPNRO2Don't know but between 50 and 499    0.033404114 0.0232499766
## MPNRO2500 or more                          0.128853612 0.0148288868
## new_MARSTADivorced                         0.058736532 0.0171760497
## new_MARSTAMarried                          0.055448064 0.0094543158
## INDE07MAgriculture, forestry and fishing  -0.018376968 0.0632075627
## INDE07MEnergy and water                    0.169203697 0.0296419911
## INDE07MManufacturing                       0.102229644 0.0155157579
## INDE07MConstruction                        0.174390187 0.0225424207
## INDE07MDistribution, hotels and restaurants -0.030962240 0.0124965536
## INDE07MTransport and communication         0.148729700 0.0171128999
## INDE07MBanking and finance                 0.100527666 0.0131197524
## INDE07MOther services                     -0.028134161 0.0222887708
## NSECMJ20Lower managerial and professional -0.214972397 0.0125205321
## NSECMJ20Intermediate occupations          -0.453634835 0.0141048643
## NSECMJ20Lower supervisory and technical   -0.425015381 0.0200819205
## NSECMJ20Semi-routine occupations          -0.552923764 0.0160407046
## NSECMJ20Routine occupations               -0.590493898 0.0174408853
## NSECMJ20Never worked                      -0.463865298 0.0322067267
## GOR9DYorkshire and The Humber             -0.155682125 0.0222713437
## GOR9DSouth West                           -0.123554215 0.0222382017
## GOR9DWest Midlands                        -0.110846706 0.0240530451
## GOR9DSouth East                           -0.089018724 0.0216909469
## GOR9DEast of England                      -0.075017907 0.0237355383
## GOR9DEast Midlands                        -0.153178997 0.0229611700
## GOR9DNorth East                           -0.180706389 0.0254060203
## GOR9DNorth West                           -0.137984041 0.0225070443
## GOR9DWales                                -0.170684127 0.0253828135
## GOR9DNorthern Ireland                     -0.166344748 0.0229327215
## GOR9DScotland                             -0.110272913 0.0233079479
```

```
##                                                t value      Pr(>|t|)
## (Intercept)                                104.0376850  0.000000e+00
## SEXFemale                                   -8.2388523  1.738704e-16
## AGE                                         13.5714885  5.910932e-42
## HIQUL22DHigher education                    -8.0821026  6.365952e-16
## HIQUL22DA level or equivalent              -14.6456640  1.435916e-48
## HIQUL22DGCSE A*-C or equivalent            -15.2956386  8.175496e-53
## HIQUL22DOther qualitfication                -9.2540134  2.162180e-20
## HIQUL22DNo Qualification                   -12.7596952  2.752761e-37
## FTPTPart-time                               -2.7562659  5.846546e-03
## JOBTYPNot permanent in some way             -1.2005708  2.299177e-01
## MPNRO211-19                                  0.6119073  5.405991e-01
## MPNRO220-24                                  0.6018320  5.472859e-01
## MPNRO2Don't know but under 25               -1.1455869  2.519661e-01
## MPNRO225-49                                  0.8754585  3.813245e-01
## MPNRO250-249                                 3.3217818  8.944462e-04
## MPNRO2250-499                                4.8078839  1.525363e-06
## MPNRO2Don't know but between 50 and 499      1.4367375  1.507926e-01
## MPNRO2500 or more                            8.6893651  3.644707e-18
## new_MARSTADivorced                           3.4196764  6.269566e-04
## new_MARSTAMarried                            5.8648416  4.495617e-09
## INDE07MAgriculture, forestry and fishing    -0.2907400  7.712502e-01
## INDE07MEnergy and water                      5.7082433  1.141482e-08
## INDE07MManufacturing                         6.5887625  4.435075e-11
## INDE07MConstruction                          7.7360896  1.025212e-14
## INDE07MDistribution, hotels and restaurants -2.4776623  1.322462e-02
## INDE07MTransport and communication           8.6910869  3.589883e-18
## INDE07MBanking and finance                   7.6623143  1.826121e-14
## INDE07MOther services                       -1.2622572  2.068563e-01
## NSECMJ20Lower managerial and professional  -17.1695896  4.485805e-66
## NSECMJ20Intermediate occupations           -32.1615880 6.082501e-227
## NSECMJ20Lower supervisory and technical    -21.1640805  2.047165e-99
## NSECMJ20Semi-routine occupations           -34.4700421 2.255790e-260
## NSECMJ20Routine occupations                -33.8568764 2.874888e-251
## NSECMJ20Never worked                       -14.4027458  4.972793e-47
## GOR9DYorkshire and The Humber               -6.9902439  2.744087e-12
## GOR9DSouth West                             -5.5559445  2.761146e-08
## GOR9DWest Midlands                          -4.6084271  4.057265e-06
## GOR9DSouth East                             -4.1039575  4.061421e-05
## GOR9DEast of England                        -3.1605732  1.574590e-03
## GOR9DEast Midlands                          -6.6712192  2.536871e-11
## GOR9DNorth East                             -7.1127389  1.137624e-12
## GOR9DNorth West                             -6.1307047  8.749069e-10
## GOR9DWales                                  -6.7243975  1.763205e-11
## GOR9DNorthern Ireland                       -7.2535982  4.058428e-13
## GOR9DScotland                               -4.7311292  2.232744e-06
```

Thoughts

- As expected, it didn't make much difference
- Once again thought, it is worth checking because it might alter our conclusions
- There may also be some clustering by geography e.g., different cities/regions
- We have very broad regions in our data which we could consider utilising
- We have controlled for regional differences

Some causes of violating this assumption

- Misspecification
- Data has a hierarchical structure
- Repeated observation of the same unit(s) over time
- Geographical relationships in the data

Testing

- It goes beyond the scope of this course to get into testing for violations of the assumption of independence
- At the start of the course, I asked you all to pick cross-sectional data where this would be less likely to be an issue
- You should think about whether there are likely to be any structure in your data which may cause problems
- You can at least mention this as a potential limitation and/or area for future work

Assumption 5

- The fifth assumption is quite abstract and doesn't really require us to do anything
- Suppose we have the following theoretical model:

$$y_i = \beta_0 + \beta_1 \cdot x_{1i} + u_i$$

- We run our OLS model and get estimates of everything

Assumption 5

- Suppose we were wrong and in fact, unbeknownst to us, the correct theoretical model is

$$y_i = \beta_0 + \beta_1 \cdot x_{1i} + \beta_2 \cdot x_{2i} + u_i$$

- If x_1 and x_2 are uncorrelated, the we avoid omitted variable bias and our estimate of β_1 will be unbiased
- However, our estimate of β_0 will be biased as it will include the mean of the excluded x_2
- We almost never interpret the intercept of a model, so we don't really care about this
- Including β_0 in the model ensures that our estimates of u will have zero mean

Thoughts on our model

- We have now estimated our model, tested it, and refined it
- Although not perfect, I think it is relatively robust
- We started out trying to estimate the wage gap
- In our final model, we found that women were paid about 11% less per hour than men
- This difference cannot be explained by any of the factors we controlled for

Building a robust model

- We have now discussed the things we need to think about to build a robust model
- We have seen how to build a model and have discussed the assumptions underpinning it
- I want to recap some of these now and offer some additional advice

Building a robust model

- General to specific
  - Start with all variables and whole sample
  - Reduce and refine as necessary
- Specific to general
  - Start with few variables and specific sample
  - Expand and refine incrementally
- Different disciplines and researchers have different opinions about which is appropriate
- Both are defensible so long as you thoroughly test your model

A general to specific approach to modelling

1. Theory
2. Design ideal regression model
3. Data collection
4. General model
5. Diagnostic checks and refinement

6. Specific model
7. Revise theory?
8. Present final model

Theory

- Start with a theory; could be from quantitative or qualitative research
- What are the most plausible mechanisms at play?
- What causes one variable to change in response to another
- Try to cater for all possible determinants
- Try to identify specific hypotheses to test

A general model

- Attempt your first regression model
- Make obvious modifications before starting the diagnostic/refinement process
- Examine residual plots
- Perform diagnostic tests e.g. RESET, heteroscedasticity

Potential problems

- Wrong signs
  - Check for multicollinearity
  - Try non-linear transformations
  - Consider creating interaction terms
- Unreasonably low $\bar{R}^2$
  - Transform variables
  - Drop irrelevant variables
  - Consider adding variables

Your specific model

- Should be well behaved:
  - Stable
  - Passes specifications tests; if possible
- Coefficients should be meaningful:
  - Do they make sense?
  - How do they relate to your theory/intuition?
  - Are there alternative explanations/interpretations
- Do you need to revise your theory after seeing the results?

- Sometimes it is only when you begin the empirical work that you will properly grasp your theory

Getting to your final model

- Arriving at a robust model is part art and part science
- It typically takes a lot of practice to get good at this process
- Try running different regressions, and looking at the different diagnostics
- Remember that formulating the model in the first place is probably the most important step
- Whenever you are going to present a model, you need to make sure it is robust
- Outline what testing has been done, and what problems might remain
- There are a lot of good guides online about how to check the robustness of a model
- There are also many good books on this subject

Beyond OLS

- You can spend your whole life learning new quantitative methods
- We have given a solid introduction in this course which should give you a good basis for learning new methods
- I want to finish by mentioning a some other popular regression methods
- I **DO NOT** require you to use these in the assignment
- You **WILL NOT** be given extra marks for using these methods and I **DO NOT** want to see them in your assignment

Count data

- In the first lecture, we said that counts of things were an example of a discrete variable
- We said that as a rule of thumb, if the mean of the count was less than 10, then we shouldn't use OLS
- When the count is below 10, an alternative is to use either a Poisson or Negative Binomial model
- These models explicitly account for the count nature of the data
- They don't predict infeasible values (e.g., negative values)
- They give the correct standard errors and hence allow robust hypothesis testing

Binary dependent variable

- Suppose you have a situation where an event occurs or does not, or something is true or is not
- For instance, does someone own a car?
- We might want to investigate what factors determine car ownership
- For this, we could consider a probit or logistic regression model

Categorical variables

- Sometimes out dependent variable may be categorical e.g., mode of transport taken to work
- A standard way to model this is using a multinomial logistic regression model, sometimes referred to as the multinomial logit model
- Variations on this model exist to deal with special cases e.g., nested logit model

Ordinal variables

- Ordinal variables can be treated as categorical and analysed with the multinomial logit model
- Under certain assumptions, more efficient estimation can be achieved with the ordered logit or ordered probit model

Panel data

- Sometimes we may have repeated observation of the same cross-sectional units
- Analysing such data with OLS can sometimes work, although usually doesn't
- An alternative is to use a panel data model (e.g., fixed effects or random effects)
- Such models have attractive features such as allowing us to account for unobserved heterogeneity
- They also allow robust hypothesis testing with this form of data
- Different versions exist for different dependent variable types e.g., linear, logistic and so on.

Time series data

- Sometimes we collect data which varies over time e.g., GDP, Unemployment, Inflation and so on.
- Sometimes we can use OLS to model these, but specialist models also exist
- For univariate time series, we have models such as the Autoregressive Integrated Moving Average (ARIMA) model, Generalised Autoregressive Conditional Heteroscedasticity (GARCH) model

- For multivariate analysis, we have models such as Vector Autoregressive (VAR) models and Vector Error Correction Models (VECM)

Spatial econometrics

- This class of models can be used when data are spatially referenced
- Some of the models are very similar to OLS regression, but account for the violation of the assumption of independence caused by spatial proximity
- Some models in this tradition are the Spatial Error Model, Spatial Lag Model, and the Spatial Durbin Model

The end

- You now have a strong foundation in quantitative data analysis
- There are many more methods, approaches and opinions out there
- Even if you never use the methods, you should now be able to understand and critique the work of others